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PEE 413 — Past Questions & Materials
by Renance

Course: PEE 413
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — PEE 413 2019 PAST QUESTION AND ANSWERS

Level 400 • First Semester • PastQuestion

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University of Petroleum Resources, Effurun
Department of Petroleum Engineering
PNG 413 – Natural Gas Engineering I
2019 Final Exam

QUESTION 1

A reservoir having the properties listed below was put into production and the reservoir pressures and their associated cumulative gas production were as shown in the table below.

Reservoir Pressure (psia)	Cumulative Gas Production (Gp) (MMM SCF)
1885	6.873
1620	14.002
1205	23.687
888	31.009

Reservoir porosity = 23%; Connate water saturation = 17%
Initial reservoir pressure = 2100 psia; Initial reservoir temperature = 185 °F
Initial gas formation volume factor = 0.00389; Reservoir fluid SG = 0.740,
Pseudo critical pressure = 654; Pseudo critical temperature = 380
Reservoir thickness = 150 ft;

- Employ the material balance graphical method to determine the initial gas in place (Gi)
- From your plot give the equation representing the production dynamics of the reservoir.
- What is the total area of the reservoir in acres?

QUESTION 2

To prevent condensate dropping, a pressure maintenance programme with gas injection was planned for FUPRE condensate gas reservoir in OML34 with an average pressure and temperature of 2150 psia and 187 °F respectively. The 5,000 ft vertical well was completed with a 3 inches ID tubing with roughness $\epsilon = 0.0006$ inches. The injection well head tubing pressure will be maintained at 2050 psia. Using the rewritten R. V. Smith equation by Moody et.al equation and the provided Moody friction chart determine the bottomhole injection pressure? The gas viscosity is 0.17 cp, with 0.6 gravity, pseudocritical pressure = 642psia, pseudocritical temperature = 446 °R will be injected at 7 mm scf/day

QUESTION 3

- Briefly explain why the same rock will give higher permeability to gas than oil or liquid
- Explain how it resolves the issue of the apparent higher gas flow than liquid flow exhibited by the same rock
- Determine the absolute permeability of a gas reservoir rock whose sample has the following dimensions that was used for permeability determination experiment in the laboratory. Width = 1.510 cm Height = 1.570 cm, Length = 2.1cm. The gas viscosity $\mu_g = 0.191$, and downstream pressure $P_2 = P_{sc} = 1$ atm

The results were as follows:

Run	q_g (cc/sec)	P_1 (atm)	$1/P_m$ (1/atm)
a	4.05	1.13	0.94
b	17.90	1.50	0.80
c	34.56	1.86	0.70
d	61.79	2.33	0.60

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Mat. No. _____ Date: __, 2019. __

Time: _____

Basic Instructions:

1. Duration – 3 hours
2. Answer question no. 1 and four(4) others
3. Some basic equations and constants are provided below
4. 13825 dynes/cm = 1lbm-ft/s²

$$Q = 200,000 \left[\frac{d^3}{T_a T_b} (P_2^2 - e^2 P_1^2) \frac{S}{e^2 - 1} \right]$$

$$P_1^2 - e^2 P_2^2 = \pm 6.67 \times 10^{-4} \frac{(Q T_a Z_a)^2 (e^2 - 1) f}{d^5}$$

$$P_{wf}^2 = e^S P_{TH}^2 + R^2$$

$$R^2 = (F_{Dnew} Q)^{1.935} (e^2 - 1)$$

$$F_{Dnew} = F_D \left(\frac{T_z}{540} \right)^{1.0336} \left(\frac{.72 \mu}{.0109 \gamma} \right)^{.0356}$$

$$f = 30.9208 \times 10^{-3} \frac{Q^{-.065} d^{-.056} \gamma^{-.065}}{\mu^{-.065}}$$

$$F_D = 65.710 / d^{2.614}$$

$$P_{wf}^2 = e^S P_{TH}^2$$

$$\gamma_f = \gamma_g + \frac{\frac{459 \gamma_L}{GLR}}{1 + \frac{1123}{GLR}}$$

$$v_{K'} = 17.6 \frac{[\sigma(\rho_l - \rho_x)]^{0.25}}{\rho_x^{0.5}}$$

$$q_K = \frac{3.06 p v_{K'} A}{Z T}$$

$$s = 0.0375 \frac{G h}{T_a Z_a}$$

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QUESTION 4

- a. With the aid of a typical hydrocarbon phase diagram
- Identify, sketch out and define the five(5) main types of reservoirs known to the oil and gas industry
 - With their respective surface and reservoir pressure/temperature profiles, describe and differentiate each of the types of reservoirs
- b. Define liquid loading and condensate dropping
- c. Determine the gas flow velocity and capacity that will prevent liquid loading in an 9230 ft well completed with tubing grade J-55 with weight = 9.5 lbm/ft having ID = 3½" and OD = 4½". The well flows a 500 GLR gas at tubing pressure of 1200 psi. The gas and liquid gravities are respectively 0.72 and 0.81. The fluid average viscosity is 0.0185 cp with surface tension of 68 dynes/cm. The pseudocritical pressure = 600psia, pseudocritical temperature = 450 °R and flowing tubing temperature = 192 °F

QUESTION 5

A newly discovered gas-condensate reservoir has an average bulk volume of 680 acre-ft and reservoir pressure and temperature of 3,000 psia and 230 °F respectively. The Porosity is 30 percent and the interstitial water saturation is 12 percent. The test flow parameters and other calculated properties are as follows;

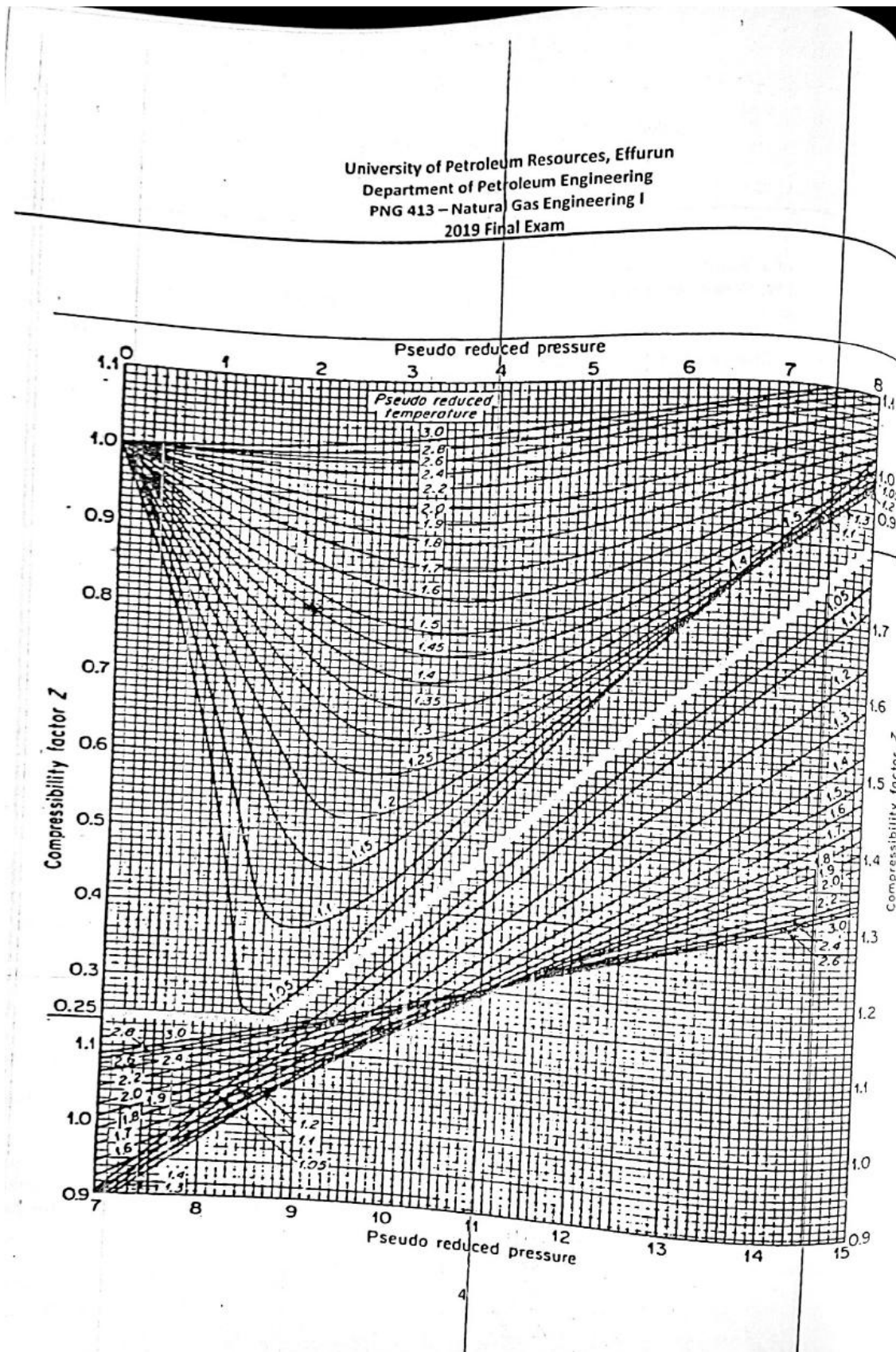
Tank condensate production 450 bbl/day; Tank condensate gravity 50° API
Tank gas production = 250 MCF per day; Tank gas gravity 1.15 (air = 1)
Separator gas production = 4,500 MCF/day; Separator gas gravity 0.7 (air = 1)
Pseudocritical temperature(Tpc) = 455; Pseudocritical pressure (Ppc) = 640
Pound-moles in one barrel of condensate (N_b) = 2.12 moles/bbl.cond
Gas mole fraction = 0.922; Condensate fraction = 0.078
Reservoir fluid gravity to Average separator gas gravity ratio = 1.40

- Determine the
 - Average gas gravity
 - Condensate-gas ratio(CGR)
 - Reservoir fluid gravity
 - Reservoir fluid compressibility factor
- Calculate the hydrocarbon volume and number of moles in this reservoir
- What is total surface volume of the gas in the reservoir
- With the given gas mole fraction, condensate mole fraction and pound-mole in one condensate barrel, calculate the actual
 - Gas-in-place in scf
 - Condensate-in-place in bbl.

QUESTION 6

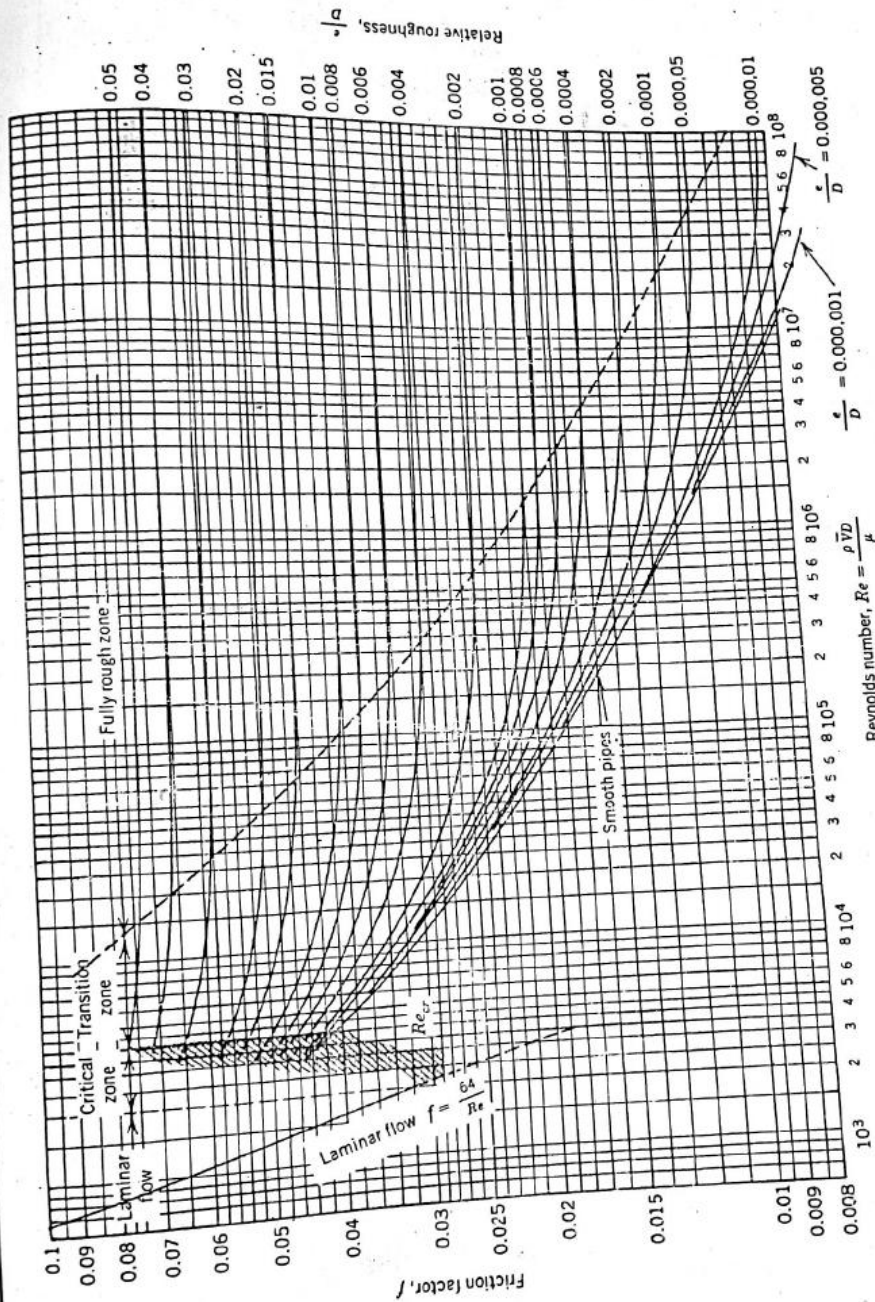
An 8,500 ft reservoir sandstone has a cap rock of 500ft on top of it. The pore pressure gradient and overburden gradient of the area are 0.462 psi/ft and 1.1 psi/ft respectively.

- Provide the lithology vertical profile sketch assuming the cap rock top is overlaid with one layer of top soil and three(3) alternating layers of shale and marble formations.
- Determine the specific gravity of the mud required to drill through this reservoir.
- A gas well was completed with a 3½" tubing(ID = 3.25") whose weight and grade are 9.3 lbm/ft and H-40, respectively. The flowing tubing pressure and temperatures are $T = 190^\circ\text{F}$ and $P = 750$ psi. The gas flowing properties are as follows; GLR = 520, $\sigma = 0.0063$ lbm-ft/s², $\rho_g = 62.4$ lbm/ft³, $\gamma_g = 0.61$ $Z = 0.962$. Determine the gas critical velocity and flow rate that will prevent liquid loading.



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Solution To Questions.Question 1Given $P_{pc} = 654$, $T_{pc} = 380$.

Reservoir Pressure (psia)	Cum Gas Production (G_p) (MMscf)	$P_r = \frac{P}{P_{pc}}$	$T_r = \frac{T}{T_{pc}}$	$z(P_r, T_r)$	P/z
2100	0	3.2110	1.6974	0.86	2441.86
1885	6.873	2.8823	1.6974	0.86	2191.86
1620	14.002	2.4771	1.6974	0.87	1862.07
1205	23.687	1.8425	1.6974	0.90	1338.89
888	31.009	1.3578	1.6974	0.91	975.82

Plotting P/z against G_p . (At the Back)

From graph

a) $G_i = 51 \text{ MMscf}$

b) The equation representing the production dynamics of the reservoir

$$\frac{P}{Z} = \text{Slope } G_p + \frac{P_i}{Z_i}$$

$$\frac{P_i}{Z_i} = \text{initial } \frac{P}{Z} = 2441.86$$

From Graph

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2441.86 - 975.82}{(0 - 32.00) \times 10^9} = -4.58 \times 10^{-8}$$

So therefore, Equation representing the Production dynamics of the reservoir is:

$$P/z = -4.58 \times 10^{-8} G_p + 2441.86$$

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© Total area of the reservoir in acres?
Using

$$G_i = \frac{43560 A h \phi (1 - S_w)}{B_{gi}}$$

$$G_i = \frac{43560 \frac{\text{ft}^3}{\text{ac ft}}}{0.00389 \frac{\text{ft}^3}{\text{scf}}} \times A \times 150 \text{ ft} \times 0.23 (1 - 0.17)$$

where $G_i = 51 \times 10^9 \text{ scf}$

$$51 \times 10^9 \text{ scf} = \frac{43560 \text{ ft}^3}{\text{ac ft}} \times A \times 150 \text{ ft} \times 0.23 (1 - 0.17)$$

$$51 \times 10^9 \text{ scf} \times 0.00389 \frac{\text{ft}^3}{\text{scf}} = \frac{43560 \text{ ft}^3}{\text{ac}} \times 28.635 A$$

$$198390000 \text{ ft}^3 = \frac{43560 \text{ ft}^3}{\text{ac}} \times 28.635 A$$

$$A = \frac{198390000 \text{ ft}^3}{\frac{43560 \text{ ft}^3}{\text{ac}} \times 28.635} = \frac{198390000}{43560 \times 28.635}$$

$$A = \frac{198390000}{1247340.6} = 159.05 \text{ ac-re}$$

$$A = \underline{\underline{159.05 \text{ acres.}}}$$

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Question 2.

From the question.

$$T_a = 187^\circ \text{F}$$

$$P_a = 2150 \text{ psia}$$

$$h = 5000 \text{ ft} = L$$

$$d = 3 \text{ inch}$$

$$P_w = 2050 \text{ psia}$$

$$T_{pc} = 446^\circ \text{R}$$

$$P_{pc} = 642 \text{ psia}$$

$$\mu = 0.17 \text{ cp}$$

$$\gamma_g = 0.6$$

$$Q = 7 \text{ mm Sec/day}$$

$$P_{wf} = ?$$

The Re-written R-V Smith Equation by Moody et al is,

$$P_{wf}^2 = e^s P_w^2 + \frac{6.67 \times 10^{-4} (Q T_a Z_a)^2 (e^s - 1) f}{d^5}$$

But for the question, Since gas injection (-ve) we have,

$$P_{wf}^2 = e^s P_w^2 - \frac{6.67 \times 10^{-4} (Q T_a Z_a)^2 (e^s - 1) f}{d^5}$$

$$Z = ?$$

$$Z(T_{pr}, P_{pr})$$

$$T_{pr} = \frac{T}{T_{pc}} = \frac{187 + 460}{446} = \frac{647}{446} = 1.45$$

$$P_{pr} = \frac{P}{P_{pc}} = \frac{2150}{624} = 3.45$$

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From Z table
 $Z(1.45, 3.45) = 0.74$
 $S = \frac{0.0375 \gamma_g L}{T_a Z_a} = \frac{0.0375 \times 0.6 \times 5000}{647 \times 0.74}$

$$S = 0.235$$

$$f = \frac{0.0175}{d^{0.224}} = \frac{0.0175}{(3)^{0.224}}$$

$$f = 0.0137$$

Substituting all into the (-ve) gas injection re-written R.V. Smith Equation by Moody et al

$$P_{wf}^2 = e^{0.235} \times 2050^2 - 6.67 \times 10^{-4} \frac{(7 \times 647 \times 0.74)^2 (e^{0.235} - 1) \times 0.0137}{(3)^5}$$

$$P_{wf}^2 = 5315778.989$$

$$P_{wf} = 2305.59 \text{ psia}$$

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Question 3.

3a) Briefly Explain why the Same rock will give higher permeability to gas than oil or Liquid.

Ans

The reason is because of the Slippage phenomenon which occurs along the pore walls resulting in higher flowrate (High permeability) in the Laboratory due to low flowing gas pressure but negligible for gas flow at reservoir Condition.

3b) Briefly Explain how it resolves the issue of the apparent higher gas flow than Liquid flow exhibited by the Same rock.

Ans

Because of the Slippage that occurs along the Pore walls, gas will have higher permeability which is not the absolute permeability of the well but the apparent permeability which varies with gas flow mean pressure, Size of gas molecules and property of the Porous medium. The relationship between the apparent permeability to gas and the absolute permeability to non reactive Liquid is given as;

$$K_a = K_L \left(1 + \frac{b}{P_m} \right)$$

3c) Solution.

Using $Q_a = \frac{K_a A (P_1^2 - P_2^2)}{2 \mu L P_{sc}}$

$$K_a = \frac{Q_a \cdot 2ALP_{sc}}{ACP_1^2 - P_2^2}$$

Given: $\mu_g = 0.191$, width = 1.510 cm, Height = 1.570 cm
 , length = 2.1 cm, $P_2 = P_{sc} = 1 \text{ atm}$

$$A = w h$$

$$= 1.510 \times 1.570$$

$$A = 2.3707$$

For Run 1.

$$K_{a1} = \frac{4.05 \times 2 \times 0.191 \times 2.1 \times 1}{2.3707 (1.13^2 - 1^2)}$$

$$K_{a1} = 4.9492 \text{ Darcy}$$

For Run 2.

$$K_{a2} = \frac{17.90 \times 2 \times 0.191 \times 2.1 \times 1}{2.3707 (1.5^2 - 1^2)}$$

$$K_{a2} = 4.8456 \text{ Darcy}$$

For Run 3.

$$K_{a3} = \frac{34.56 \times 2 \times 0.191 \times 2.1 \times 1}{2.3707 (1.86^2 - 1^2)}$$

$$K_{a3} = 4.7546 \text{ Darcy}$$

For Run 4

$$K_{a4} = \frac{61.79 \times 2 \times 0.191 \times 2.1 \times 1}{2.3707 (2.33^2 - 1^2)}$$

$$K_{a4} = 4.7209 \text{ Darcy}$$

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Question 3 Contd.

Table

Run	q_g (cc/sec)	P_1 (atm)	$1/P_m$ (atm)	K_a (darcy)
a	4.05	1.13	0.94	4.9492
b	17.90	1.50	0.80	4.8456
c	84.56	1.86	0.70	4.754
d	61.79	2.33	0.60	4.7209

From Graph $K_L = 4.36$ Darcy $\therefore K_a$.

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4.95 - 4.72}{0.9 - 0.6} = 0.7667$$

$$\text{Slope} = 0.7667$$

From

$$\text{Slope} = K_L \cdot b$$

$$b = \frac{\text{Slope}}{K_L}$$

$$b = \frac{0.7667}{4.36}$$

$$b = \underline{\underline{0.1758}}$$

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Therefore

$$K_g = K_L \left(1 + \frac{5}{P_m} \right)$$

Taking mean pressure P_m as 1 atm
Absolute Permeability

$$K_g = 4.36 \left(1 + \frac{0.1758}{1} \right)$$

$$K_g = 4.36 \times 1.1758$$

$$K_g = \underline{\underline{5.1265 \text{ darcy}}}$$

K_g for this question = 4.36 darcy Since this
the permeability of gas was used.

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Question 4.

4ai)

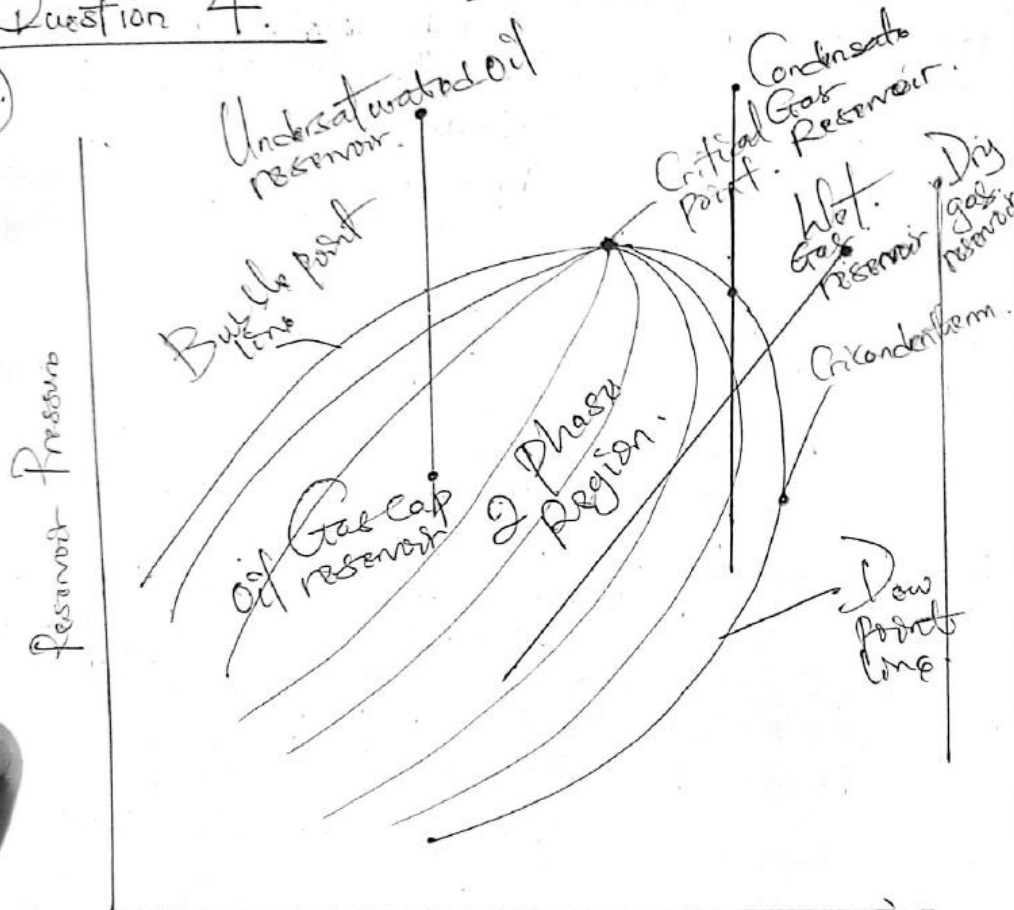


Diagram showing five (5) types of reservoirs (max).
 For oil reservoirs the reservoir temperature is less than the Critical point. $T_r < T_c$. (Temperature).
 For gas reservoirs the reservoir temperature is greater than the Critical temperature point. $T_r > T_c$.
 - In differentiating between the different types of Oil reservoirs the reservoir pressure comes into play.

4aii) Dry Gas Reservoir

Has Reservoir and Separator P & T Conditions above the dew point

• Wet Gas Reservoir

Has Reservoir P & T Conditions above dew point while Separator Condition in 2-Phase Region

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- Condensate Gas Reservoir.
Has Reservoir p & T Conditions above dew point and between Critical point and Critical temperature.
- Undersaturated Oil Reservoir.
Has its reservoir Pressure greater than the Bubble point pressure and it's also less than the Critical point.
- Gas Cap Oil Reservoir.
Has its reservoir pressure less than the bubble point pressure leading to a 2 phase mixture in the two phase Region.
- 4b) • Liquid Loading is the falling back of condensed liquid droplets in the gas stream that causes accumulation of Liquid (Condensate and/or water) at gas wellbore bottom due to the natural gas production velocity dropping below certain level referred to as "Gas Critical Velocity".
- Condensate dropping is a phenomenon that occurs at the wellbore. Condensate liquid can be dropped-out due to change in fluid pressure and temperature. If the gas doesn't have sufficient energy to carry liquid to the surface liquid loading occurs.

Also refer to your Pdf Under Condensates to Understand more on Condensate dropping, Condensate blocking & Liquid Loading to better enhance your definition.

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$$V_g = \frac{17.6 \cdot [6(P_L - P_g)]^{0.25}}{P_g^{0.5}}$$

$$V_g = \frac{17.6 \cdot [68(50.544 - 5.895)]^{0.25}}{(5.895)^{0.5}}$$

$$V_g = 53.8085 \text{ ft/s.}$$

$V_g = \text{gas flow velocity}$

To Calculate the gas Capacity, you use;

$$q_g = \frac{3.06 \cdot P \cdot V_g \cdot A}{Z \cdot T_R}$$

$$q_g = \frac{3.06 \times 1200 \times 53.8085 \times 1386}{0.8 \times 652}$$

$$q_g = 57.3810.4092 \text{ mmscf/day}$$

$$q_g = 573.8 \text{ mmscf/day}$$

Note:- γ_g is the real gas gravity that will be calculated before using it to find the gas density and ultimately calculating your gas velocity.

$$\gamma_g = 0.72, GUR = 500$$

$$\gamma_L = 0.81$$

$$\gamma_f = \gamma_g + \frac{459 \gamma_L}{GUR}$$

$$\gamma_f = 0.72 + \frac{459(0.81)}{500}$$

$$\gamma_f = 0.9491$$

$$P_g = \frac{2.7 P \gamma_f}{T_R Z}$$

$$P_g = \frac{2.7 \times 1200 \times 0.9491}{652 \times 0.8}$$

$$P_g = 5.895$$

$$\gamma_L = \frac{P_L}{P_w} \quad P_w = 62.4 \text{ lb/ft}^3$$

$$P_L = P_w \times \gamma_L$$

$$P_L = 62.4 \times 0.81$$

$$P_L = 50.544.$$

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Question 4 Contd.

Solution.

$$ID = 3\frac{1}{2}" = 3.5" = 3.5 \times 12 = 42 \text{ ft}$$

$$A = \frac{\pi (ID)^2}{4}$$

$$A = \frac{\pi (42)^2}{4}$$

$$A = 1385 \text{ ft}^2$$

$$Z(P_{pr}, T_{pr})$$

$$P_{pc} = 600 \text{ psia}, T_{pc} = 450^\circ \text{R}; P = 1200 \text{ psi}; T = 192^\circ \text{F}$$

$$P_{pr} = \frac{P}{P_{pc}}, T_{pr} = \frac{T}{T_{pc}}$$

$$P_{pr} = \frac{1200}{600}, T_{pr} = \frac{192 + 460}{450}$$

$$P_{pr} = 2, T_{pr} = \frac{652}{450}$$

$$T_{pr} = 1.45$$

$$Z(P_{pr}, T_{pr}) = Z(2, 1.45) = 0.8$$

$$Z = 0.8$$

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Question 5.

Given

$$V_b = 680 \text{ acre-ft}$$

$$P = 3000 \text{ psia}$$

$$T = 280^\circ\text{F}$$

$$\phi = 30\%$$

$$S_w = 12\%$$

Other Parameters as Stated.

a.i) Average Gas Gravity.

$$\gamma_g = \frac{V_s \gamma_s + V_T \gamma_T}{V_s + V_T}$$

$$\begin{aligned} \gamma_g &= \frac{4,500 \times 10^3 \times 0.7 + 250 \times 10^3 \times 1.15}{4,500 \times 10^3 + 250 \times 10^3} \\ &= \frac{(4,500 \times 0.7 + 250 \times 1.15) (\times 10^3)}{(4,500 + 250) (\times 10^3)} \end{aligned}$$

$$\gamma_g = 0.7237$$

a.ii) Condensate Gas Ratio (CGR)

$$\begin{aligned} \text{CGR} &= \frac{Q_c}{Q_{g\text{tank}} + Q_{g\text{trap}}} \\ &= \frac{450 \times 1000}{250 + 4500} \\ &= 94.7466 \text{ bbl/mm ft}^3 \end{aligned}$$

a.iii) Reservoir fluid Gravity

Given, Reservoir fluid gravity to Average separator gas gravity = 1.40

$$\begin{aligned} \text{Reservoir fluid Gravity} &= 1.40 \times 0.7237 \\ &= 1.0132 \end{aligned}$$

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9iv) Reservoir fluid Compressibility factor.
 $Z(T_{pr}, P_{pr})$

$$T_{pr} = \frac{T}{T_{pc}} = \frac{230 + 460}{455} = 1.516$$

$$P_{pr} = \frac{P}{P_{pc}} = \frac{3000}{640} = 4.6875$$

Using Compressibility Chart $Z = 0.8$.

b.) i) Hydrocarbon volume and no of moles in the Reservoir.

$$V_h = 43560 V_b \phi (1 - S_w) \\ = 43560 \times 680 \times 0.3 (1 - 0.12)$$

$$V_h = 7,819,891.2 \text{ ft}^3$$

$$V_h = 7.82 \text{ MMft}^3.$$

ii) $N = \frac{PV_h}{ZRT}$ (No of moles)

Where $R = 10.732$.

$$N = \frac{3000 \times 7,819,891.2}{0.8 \times 10.732 \times 690}$$

$$N = 3960064.172 \text{ lb-mol.}$$

c.) Total Surface Volume of the gas in the Reservoir

$$V_g = 379N$$

$$V_g = 379 \times 3960064.172$$

$$V_g = 1500864321.8 \text{ cf.}$$

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Question 5 Contd.

d) i) Gas in place in Sep.
 $= V_g \times \text{Gas mole fraction.}$
 $= 1500864321 \times 0.922$
 $= 1383796904 \text{ ft}^3$
 $= 1.38 \text{ mmmft}^3.$

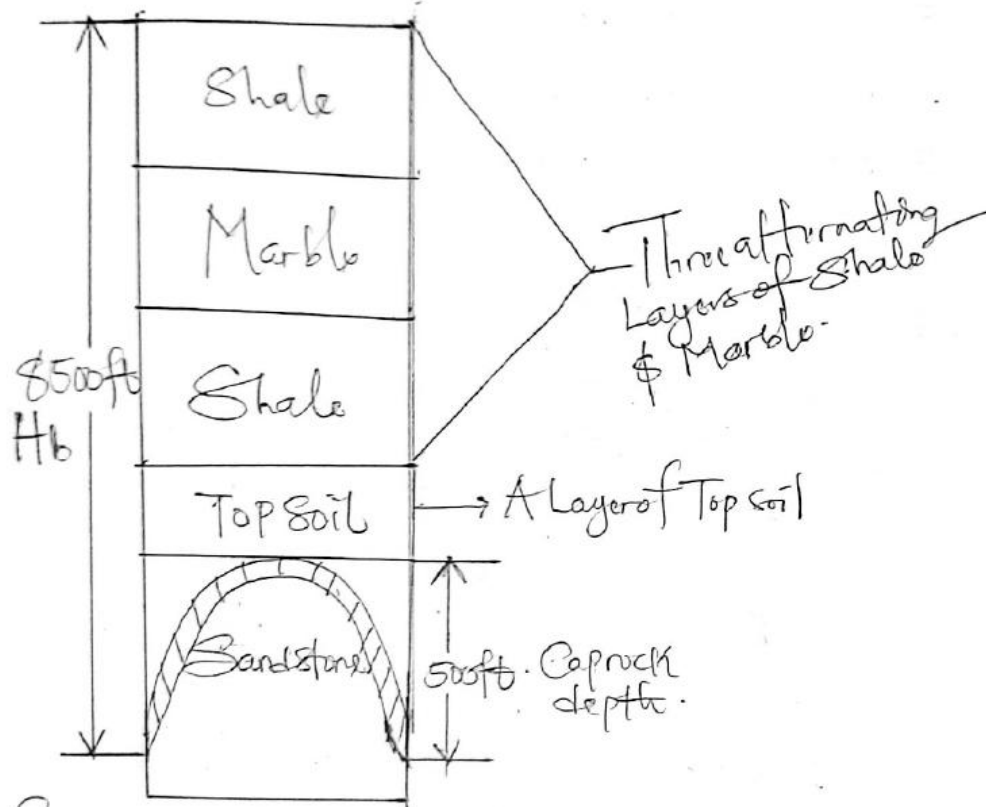
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 ii) Condensate in place in bbl
 $= \frac{\text{Condensate fraction} \times N}{NB}$
 $= \frac{0.078 \times 3960064.72}{2.12}$
 $= 145700.47 \text{ bbl}$
 $= \underline{\underline{145.7 \text{ M bbl.}}}$

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Question 6

6a) Lithology Vertical profile Sketch.



Given

Pore Pressure gradient = 0.462 psi/ft

Overburden gradient = 1.1 psi/ft

$H_a = H_b - \text{Caprock depth.}$

$= 8500 - 500$

$H_a = 8000 \text{ ft.}$

Calculating the Expected Pressure (Total)

$P_T = \text{Pore pressure gradient} \times H_a + \text{Overburden gradient} \times \text{Caprock depth.}$

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Question 6 Contd.

$$= 0.462 \text{ psi/ft} \times 8000 \text{ ft} + 1.1 \text{ psi/ft} \times 500 \text{ ft}$$

$$P_T = 3696 + 550$$

$$P_T = 4246 \text{ psi}$$

From,

$$P_T = \frac{\rho H}{144}$$

$$\rho = \frac{P_T \times 144}{H}$$

$$\rho = \frac{4246 \times 144}{8500}$$

$$\rho_{\text{mud}} = 71.93 \text{ lb/ft}^3$$

$$b) \gamma_{\text{mud}} = \frac{\rho_{\text{mud}}}{\rho_{\text{water}}}$$

$$\rho_{\text{water}} = 62.4 \text{ lb/ft}^3$$

$$\gamma_{\text{mud}} = \frac{71.93}{62.4}$$

$$\gamma_{\text{mud}} = 1.15$$

c) - Given.

$$ID = 3.25''$$

$$P = 750 \text{ psi}$$

$$T = 190^\circ \text{F}$$

$$Z = 0.962$$

$$G_{LR} = 520$$

$$\rho = 0.0063 \text{ lbm/ft}^3$$

$$\rho_L = 62.4 \text{ lbm/ft}^3$$

$$\gamma_g = 0.61$$

$$V_{gc} = ?$$

$$\rho_{gc} = ?$$

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$$A = \frac{\pi D^2}{4} = \frac{\pi \left(\frac{3.25}{12}\right)^2}{4} = 0.0576$$

$$\gamma_L = \frac{C_L}{C_w} = \frac{62.4}{62.4} = 1 \quad \text{where } C_w = 62.4 \text{ lbm/ft}^3$$

$$\gamma_L = 1, \quad \gamma_g = 0.61$$

$$\gamma_F = \gamma_g + \frac{459 \gamma_L}{G_{LR}} = \frac{0.61 + \frac{459(1)}{520}}{1 + \frac{1123}{G_{LR}}} = \frac{0.61 + \frac{459(1)}{520}}{1 + \frac{1123}{520}}$$

$$\gamma_F = 0.8894$$

$$C_g = \frac{2.7 \gamma_F}{T_{RZ}} = \frac{2.7 \times 0.8894 \times 750}{656 \times 0.962} = 2.8803$$

$$V_{gc} = \frac{17.6 [6(C_L - C_g)]^{0.25}}{C_g^{0.5}} = \frac{17.6 [0.0063(62.4 - 2.8803)]^{0.25}}{(2.8803)^{0.5}}$$

$$V_{gc} = 8.1151 \text{ ft/s}, \quad V_{ge} = 8.12 \text{ ft/s}$$

$$q_{ge} = \frac{3.06 P V_{ge} A}{T_{gZ}}$$

$$q_{gc} = \frac{3.06 \times 750 \times 8.115 \times 0.0576}{650 \times 0.962}$$

$$q_{ge} = 1.7156 \text{ mmsec/day}$$

$$q_{gc} = \underline{\underline{1.72 \text{ mmsec/day}}}$$

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